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*
*      STAAD.Pro
*      Version 2007    Build 04
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date=    JAN 18, 2013
*      Time=    16:25: 7
*
*      USER ID:
*****

1. STAAD SPACE
INPUT FILE: 20121120 T-Goal Post 039 FR2.STD
2. START JOB INFORMATION
3. ENGINEER DATE 12-MAY-12
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 2 5 1 -5.73; 4 5 4.495 -5.73; 5 5 4.495 -4.98; 6 5 4.495 -6.65
9. MEMBER INCIDENCES
10. 3 2 4; 4 4 6; 5 4 5
11. DEFINE MATERIAL START
12. ISOTROPIC STEEL
13. E 2.05E+008
14. POISSON 0.3
15. DENSITY 76.8195
16. ALPHA 1.2E-005
17. DAMP 0.03
18. END DEFINE MATERIAL
19. MEMBER PROPERTY JAPANESE
20. 4 5 TABLE ST H428X407X20
21. 3 TABLE ST H428X407X20
22. CONSTANTS
23. BETA 90 MEMB 3
24. MATERIAL STEEL ALL
25. SUPPORTS
26. 2 FIXED
27. ***** DEAD LOAD *****
28. LOAD 1 LOADTYPE NONE TITLE DL
29. SELFWEIGHT Y -1
30. MEMBER LOAD
31. 4 5 CON GY -19.6 0.4
32. ***** FRICTIONAL LOAD *****
33. LOAD 2 LOADTYPE NONE TITLE FL
34. MEMBER LOAD
35. 4 5 CON GX -5.88 0.4
36. ***** WIND LOAD X-DIRECTION*****
37. LOAD 3 LOADTYPE NONE TITLE WLX
38. MEMBER LOAD
39. 4 5 UNI GX -0.364
40. 3 UNI GX -0.364
41. ***** WIND LOAD Z-DIRECTION*****
42. LOAD 4 LOADTYPE NONE TITLE WLZ
43. MEMBER LOAD
44. 3 UNI GZ -0.364
45. ***** NOTIONAL LOAD X-DIRECTION*****
46. LOAD 5 LOADTYPE NONE TITLE NLX
47. SELFWEIGHT X -0.015
48. MEMBER LOAD
49. 4 5 CON GX -0.294 0.4
50. *****NOTIONAL LOAD Z-DIRECTION*****
51. LOAD 6 LOADTYPE NONE TITLE NLZ
52. SELFWEIGHT Z -0.015
53. MEMBER LOAD
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54. 4 5 CON GZ -0.294 0.4
55. ***** LOAD COMBINATION - SERVICE *****
56. LOAD COMB 7 1.0DL+1.0FL
57. 1 1.0 2 1.0
58. LOAD COMB 8 1.0DL+1.0FL+1.0WLX
59. 1 1.0 2 1.0 3 1.0
60. LOAD COMB 9 1.0DL+1.0FL+1.0WLX
61. 1 1.0 2 1.0 4 1.0
62. LOAD COMB 10 1.0DL+1.0FL+1.0NLX
63. 1 1.0 2 1.0 5 1.0
64. LOAD COMB 11 1.0DL+1.0FL+1.0NLZ
65. 1 1.0 2 1.0 6 1.0
66. ***** LOAD COMBINATION - ULTIMATE *****
67. LOAD COMB 12 1.0DL+1.4FL
68. 1 1.0 2 1.4
69. LOAD COMB 13 1.4DL+1.4FL
70. 1 1.4 2 1.4
71. LOAD COMB 14 1.0DL+1.4FL+1.4WLX
72. 1 1.0 2 1.4 3 1.4
73. LOAD COMB 15 1.0DL+1.4FL+1.4WLZ
74. 1 1.0 2 1.4 4 1.4
75. LOAD COMB 16 1.0DL+1.4FL+1.4NLX
76. 1 1.0 2 1.4 5 1.4
77. LOAD COMB 17 1.0DL+1.4FL+1.4NLZ
78. 1 1.0 2 1.4 6 1.4
79. PERFORM ANALYSIS

P R O B L E M   S T A T I S T I C S
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NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =      4/      3/      1

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES =      6, TOTAL DEGREES OF FREEDOM =      18

80. LOAD LIST 7 TO 11
81. PRINT JOINT DISPLACEMENTS ALL

JOINT DISPLACEMENT (CM      RADIANS)      STRUCTURE TYPE = SPACE
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JOINT  LOAD  X-TRANS  Y-TRANS  Z-TRANS  X-ROTAN  Y-ROTAN  Z-ROTAN

2      7      0.0000  0.0000  0.0000  0.0000  0.0000  0.0000
      8      0.0000  0.0000  0.0000  0.0000  0.0000  0.0000
      9      0.0000  0.0000  0.0000  0.0000  0.0000  0.0000
     10      0.0000  0.0000  0.0000  0.0000  0.0000  0.0000
     11      0.0000  0.0000  0.0000  0.0000  0.0000  0.0000
4      7      -0.2099 -0.0023 -0.0010  0.0000  0.0000  0.0009
      8      -0.2293 -0.0023 -0.0010  0.0000  0.0002  0.0010
      9      -0.2099 -0.0023 -0.0039  0.0000  0.0000  0.0009
     10      -0.2227 -0.0023 -0.0010  0.0000  0.0000  0.0009
     11      -0.2099 -0.0023 -0.0055  0.0000  0.0000  0.0009
5      7      -0.2105 -0.0036 -0.0010  0.0000  0.0000  0.0009
      8      -0.2162 -0.0036 -0.0010  0.0000  0.0002  0.0010
      9      -0.2105 -0.0029 -0.0039  0.0000  0.0000  0.0009
     10      -0.2216 -0.0036 -0.0010  0.0000  0.0000  0.0009
     11      -0.2105 -0.0023 -0.0055  0.0000  0.0000  0.0009
6      7      -0.2105 -0.0047 -0.0010  0.0000  0.0000  0.0009
      8      -0.2467 -0.0047 -0.0010  0.0000  0.0002  0.0010
      9      -0.2105 -0.0056 -0.0039  0.0000  0.0000  0.0009
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11	-0.2105	-0.0063	-0.0055	0.0000	0.0000	0.0009
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***** END OF LATEST ANALYSIS RESULT *****

82. PRINT SUPPORT REACTION ALL

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
2	7	11.76	53.51	0.00	0.39	0.00	-41.10
	8	13.64	53.51	0.00	0.39	-0.05	-45.45
	9	11.76	53.51	1.21	2.51	0.00	-41.10
	10	12.56	53.51	0.00	0.39	-0.01	-43.65
	11	11.76	53.51	0.00	2.94	0.00	-41.10

***** END OF LATEST ANALYSIS RESULT *****

83. LOAD LIST 12 TO 17

84. PRINT MEMBER FORCES ALL

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
3	12	2	53.51	0.00	16.46	0.00	-57.54	0.39
		4	-43.83	0.00	-16.46	0.00	0.00	-0.39
	13	2	74.92	0.00	16.46	0.00	-57.54	0.55
		4	-61.36	0.00	-16.46	0.00	0.00	-0.55
	14	2	53.51	0.00	19.10	-0.07	-63.63	0.39
		4	-43.83	0.00	-17.32	0.07	0.00	-0.39
	15	2	53.51	1.69	16.46	0.00	-57.54	3.35
		4	-43.83	0.00	-16.46	0.00	0.00	-0.39
	16	2	53.51	0.00	17.59	-0.01	-61.11	0.39
		4	-43.83	0.00	-17.38	0.01	0.00	-0.39
	17	2	53.51	1.12	16.46	0.00	-57.54	3.97
		4	-43.83	-0.92	-16.46	0.00	0.00	-0.39
4	12	4	0.00	22.15	8.23	0.00	-3.29	9.01
		6	0.00	0.00	0.00	0.00	0.00	0.00
	13	4	0.00	31.01	8.23	0.00	-3.29	12.62
		6	0.00	0.00	0.00	0.00	0.00	0.00
	14	4	0.00	22.15	8.70	0.00	-3.51	9.01
		6	0.00	0.00	0.00	0.00	0.00	0.00
	15	4	0.00	22.15	8.23	0.00	-3.29	9.01
		6	0.00	0.00	0.00	0.00	0.00	0.00
	16	4	0.00	22.15	8.70	0.00	-3.48	9.01
		6	0.00	0.00	0.00	0.00	0.00	0.00
	17	4	-0.47	22.15	8.23	0.00	-3.29	9.01
		6	0.00	0.00	0.00	0.00	0.00	0.00
5	12	4	0.00	21.68	-8.23	0.00	3.29	8.62
		5	0.00	0.00	0.00	0.00	0.00	0.00
	13	4	0.00	30.35	-8.23	0.00	3.29	12.07
		5	0.00	0.00	0.00	0.00	0.00	0.00

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	5	0.00	0.00	0.00	0.00	0.00	0.00
15	4	0.00	21.68	-8.23	0.00	3.29	8.62
	5	0.00	0.00	0.00	0.00	0.00	0.00
16	4	0.00	21.68	-8.69	0.00	3.47	8.62
	5	0.00	0.00	0.00	0.00	0.00	0.00
17	4	0.46	21.68	-8.23	0.00	3.29	8.62
	5	0.00	0.00	0.00	0.00	0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

85. LOAD LIST ALL

86. PARAMETER 1

87. CODE BS5950

88. BEAM 2 ALL

89. CHECK CODE ALL

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.12_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
3 ST	H428X407X20	PASS	ANNEX I.1	0.083	14
		53.51 C	63.63	0.39	-
4 ST	H428X407X20	PASS	BS-4.2.3-(Y)	0.023	13
		0.00	-3.29	12.62	0.00
5 ST	H428X407X20	PASS	BS-4.2.3-(Y)	0.022	13
		0.00	3.29	12.07	0.00

***** END OF TABULATED RESULT OF DESIGN *****

90. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= JAN 18,2013 TIME= 16:25: 9 ****

* For questions on STAAD.Pro, please contact *

* Research Engineers Offices at the following locations *

* *

	Telephone	Email
* USA:	+1 (714)974-2500	support@bentley.com
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